

Leeseok Kim

Curriculum Vitae

✉ leeseok@unm.com
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Academic Appointments

2026–Present **Postdoctoral Researcher**, *University of New Mexico*, Albuquerque, NM.
Based at Los Alamos National Laboratory (Theoretical Division, T-4).
Mentor: [Luis Pedro García-Pintos](#).

Education

2021–2026 **Ph.D., Computer Engineering**, *University of New Mexico*, Albuquerque, NM, *with Distinction*.
Dissertation: [Quantum Control Protocols for Robust Quantum Computing](#). Advisor: [Milad Marvian](#).
2017–2021 **B.A., Physics and Mathematics**, *Cornell University*, Ithaca, NY.

Research Interests

Quantum computing, Quantum simulation, Open quantum systems, Quantum control.

Publications

- 2026 [L. Kim](#), “Rapid mixing of quantum spin chains at any finite temperature.” [arXiv:2610.01190](#).
2026 [L. Kim](#) and [L.P. García-Pintos](#), “Randomized product formulas beyond optimal deterministic scaling.” [arXiv:2608.07720](#).
2026 [L. Kim](#), [F. Riberi](#), [K. Baca](#), and [M. Marvian](#), “Randomized Quantum Optimal Control.” [arXiv:2607.10946](#).
2026 [L. Kim](#) and [M. Marvian](#), “Robust high-order quantum simulation using finite-width pulses.” [arXiv:2603.15502](#).
2026 [L. Kim](#) and [M. Marvian](#), “High-order dynamical decoupling in the weak-coupling regime.” *Physical Review Letters* 137, 120801.
2026 [K. R. Dizaji](#), [L. Kim](#), [M. Marvian](#), and [C. Arenz](#), “High-order Zeno sequences.” *Physical Review Research* 8(2), 023051.
2026 [C. Yi](#), [L. Kim](#), and [M. Marvian](#), “Faster Randomized Dynamical Decoupling.” *Physical Review Letters* 136, 010601.
2024 [L. Kim](#), [S. Lloyd](#), and [M. Marvian](#), “Hamiltonian Quantum Generative Adversarial Networks.” *Physical Review Research* 6(3), 033019.
2023 [L. Kim](#) and [J. L. Palacios](#), “On the Moments of the Number of Hires in the Assistant Hiring Algorithm.” *Algorithms* 16(2), 70.
2022 [T. Hur*](#), [L. Kim*](#), and [D. K. Park](#), “Quantum convolutional neural network for classical data classification.” *Quantum Machine Intelligence* 4(1), 3.
*Equal contribution.

Selected Talks

- Faster Randomized Dynamical Decoupling
2025 **SQInT 2025**, PAIS, UNM, Albuquerque, NM.
2025 **QSim 2025 [Recorded Talk]**, IBM, New York, NY.

Hamiltonian Quantum Generative Adversarial Networks

2023 **AQC 2023**, PAIS, UNM, Albuquerque, NM.

2023 **QAISG's QML Seminar [Recorded Talk]**, Centre for Quantum Technologies, NUS, Singapore (virtual).

Patent Applications

2025 **Faster Randomized Dynamical Decoupling [Link]**, C. Yi, L. Kim, and M. Marvian, U.S. Patent Application No. 19/339,205, (Filed: Sep 2025).

Professional Service

Peer review PRX Quantum, Quantum, PRA.

(Last updated: Oct 2026)